```
}
```

#A HKDF-Extract takes a random salt value and the input key material
#B If a salt is not provided then an all-zero salt is used
#C The result it the output of HMAC using the salt as the key and the key material as the input

The HKDF key for OSCORE can then be calculated from the Master Key and Master Salt as follows:

```
var hkdfKey = HKDF.extract(masterSalt, masterKey);
```

The sender and recipient keys are then derived from this master HKDF key using the HKDF-Expand function from chapter 10, as shown in listing 13.8. A context argument is generated as a CBOR array, containing the following items in order:

- The Sender ID or Recipient ID, depending on which key is being derived.
- The ID Context parameter, if specified, or a zero-length byte array otherwise.
- The COSE algorithm identifier for the authenticated encryption algorithm being used.
- The string "Key" encoded as a CBOR binary string in ASCII.
- The size of the key to be derived, in bytes.

This is then passed to the `HKDF.expand()` method to derive the key. Create a new file named Oscore.java and copy the